

Multi-dimensional Scaling and IRT Models

Item Response Theory (IRT) models originated in the mid-20th century as a way to rigorously measure latent traits—such as ability, ideology, or attitudes—based on discrete outcomes like test responses or survey answers. Developed initially at the Educational Testing Service (ETS) for standardized testing, IRT provided a principled statistical framework to model the relationship between an individual's latent characteristics and their observed item responses. Unlike classical test theory, which focused on total scores, IRT models allow each item to have its own difficulty and discrimination parameters, enabling more precise inferences and supporting applications like adaptive testing and item calibration.

In political science, IRT models were adopted to study ideological preferences, voting behavior, and mass attitudes. The seminal Poole and Rosenthal (1985) NOMINATE framework modeled legislators' ideal points in a latent ideological space, while later work by Clinton, Jackman, and Rivers (2004) recast roll-call voting in terms of 2-parameter IRT models. More broadly, IRT has become a workhorse for measuring latent traits across survey research, public opinion, and comparative political behavior. Recent advances allow for the integration of covariates, alternative response models (e.g., hurdle, ordinal), and applications to high-dimensional settings, linking IRT to contemporary tools like text embeddings and neural network architectures.

Topics Covered in This Lesson

-  Overview of IRT as Generalized Linear Mixed Models (GLMMs)
 - 1PL (Rasch), 2PL, and 3PL formulations as random effects models
-  Derivation of the 2PL model from a latent utility framework
-  Extensions in `brms`
 - Custom link functions (logit, probit, cloglog)
 - Covariates on items and individuals
 - Non-binary outcomes (count, ordinal, hurdle)
-  3PL models as hurdle models
 - Modeling intensive/extensive response behavior separately
-  Complete example using `brms` for estimation
 - Simulation, estimation, visualization

-  Extensions using **Stan**
 - Latent trait estimation and use in structural models
 - Joint vs. two-stage approaches
-  Conceptual bridge to modern embedding models and LLMs
 - Interpreting latent traits and discrimination as attention weights
 - KQV structure as a generalization of IRT

Notation and Data

Before diving into the models, here is a quick guide to the indices, data, and parameters that will appear in the 1-, 2-, and 3-parameter IRT models below.

IRT Model Notation Overview

Symbol	Description	Type
i	Index for items (e.g., questions, bills)	Index (1,...,I)
j	Index for individuals (e.g., students, legislators)	Index (1,...,J)
y_{ij}	Binary response: 1 = correct/agree/yes; 0 = incorrect/no	Observed Data
θ_j	Latent trait of person j (ability, ideology)	Parameter
β_i	Difficulty of item i (location on trait scale)	Parameter
α_i	Discrimination of item i (steepness/sorting power)	Parameter (2PL+)
c_i	Guessing/slip parameter for item i	Parameter (3PL)

1. IRT as Random Effects Models

Many IRT models can be written as **generalized linear mixed models (GLMMs)**, with individuals and items as grouping variables and various levels of complexity:

► 1PL Model (Rasch Model)

$$\Pr(y_{ij} = 1) = \text{logit}^{-1}(\theta_j - \beta_i)$$

- θ_j : ability or latent trait of person j (random intercept)
- β_i : difficulty of item i (fixed or random intercept)
- Assumes all items discriminate equally (slope = 1)

🧠 Interpretation: This is a simple random intercept model with person-level effects and item difficulties.

➤ 2PL Model (Two-Parameter Logistic)

$$\Pr(y_{ij} = 1) = \text{logit}^{-1}(\alpha_i(\theta_j - \beta_i))$$

- Adds a discrimination parameter α_i for each item
- Can be re-expressed as a random slope model:
 - $y_{ij} \sim \text{logit}^{-1}(b_{i0} + b_{i1}\theta_j)$

🧠 Interpretation: Different items can discriminate better or worse between low- and high-ability individuals.

➤ 3PL Model (Three-Parameter Logistic)

$$\Pr(y_{ij} = 1) = c_i + (1 - c_i) \cdot \text{logit}^{-1}(\alpha_i(\theta_j - \beta_i))$$

- Introduces a **guessing parameter** c_i , modeling the probability of a correct response due to chance
- Especially relevant in multiple-choice tests or when guessing is common

🧠 Interpretation: This is akin to adding a lower asymptote to the logistic curve—people with very low ability can still get items right sometimes.

➤ Hurdle Interpretation of the 3PL Model

The 3-parameter logistic model can also be understood as a **two-part (hurdle) process**:

1. Extensive Margin (Engagement)

First, a person decides whether to *engage with the item at all*. This might reflect:

- Attention or effort (in survey data)
- Willingness to take a stand (in political roll-call votes)
- Decision to guess (in educational testing)

2. Intensive Margin (Conditional Response)

Conditional on engagement, the person answers based on their **latent trait** θ_j , item **difficulty** β_i , and **discrimination** α_i .

This reformulation motivates modeling the 3PL as a **hurdle model**:

$$\Pr(y_{ij} = 1) = \Pr(\text{engage}_{ij} = 1) \cdot \Pr(\text{correct}_{ij} = 1 \mid \text{engage}_{ij} = 1)$$

Where:

- The **first term** captures the *structural* likelihood of engagement (i.e., not skipping, not abstaining, not random guessing)
- The **second term** is the familiar 2PL logistic response

This treats zeros as arising from two processes:

- A structural zero process (non-engagement or always-wrong response)
- A conditional response process among those likely to engage



When is this useful?

- Roll-call data where abstentions and votes are mixed together
- Survey items with many nonresponses or “don’t know” answers
- Testing environments with guessing or gaming behavior



This perspective also connects naturally to zero-inflated and dropout models, often used in text analysis and online behavior modeling.

Summary Table:

Model	Parameters	Interpretation
1PL	θ_j, β_i	Random effects, equal discrimination
2PL	$\theta_j, \beta_i, \alpha_i$	Adds item-specific slopes
3PL	$\theta_j, \beta_i, \alpha_i, c_i$	Adds lower asymptote (guessing/slip)



2. Utility Derivation of the 2PL Model

The 2-parameter logistic (2PL) model can be derived from a **latent utility framework**, which assumes that individuals make binary decisions based on an underlying, unobserved preference or ability. This intuition connects IRT to models of choice and spatial voting common in political science and economics.

► Step 1: Latent Utility Setup

Suppose we model an individual j 's response to item i using a latent utility function:

$$U_{ij} = \alpha_i \theta_j - \beta_i + \varepsilon_{ij}$$

Where:

- θ_j : the **latent trait** of person j
 - In **political science**, this could represent a legislator's ideology.
 - In **psychometrics**, it might reflect a student's math ability.
 - β_i : the **difficulty** or **cutpoint** of item i
 - For example, a very hard math problem (or a bill requiring extreme ideological commitment) would have high β_i .
 - α_i : the **discrimination** parameter
 - This captures how sharply the item distinguishes between individuals with different trait levels. A high α_i means small differences in θ_j lead to large changes in response probability.
 - ε_{ij} : a random error term, typically assumed to follow a standard **logistic** distribution. Different choices on this error term lead to different types of outcomes (binary/count/continuous).
-

► Step 2: From Utility to Probabilities

We observe $y_{ij} = 1$ (e.g., correct answer, "yes" vote, agreement) if the utility exceeds zero:

$$\Pr(y_{ij} = 1) = \Pr(U_{ij} > 0) = \Pr(\alpha_i \theta_j - \beta_i + \varepsilon_{ij} > 0)$$

Assuming $\varepsilon_{ij} \sim \text{Logistic}(0, 1)$, this simplifies to:

$$\Pr(y_{ij} = 1) = \text{logit}^{-1}(\alpha_i(\theta_j - \beta_i))$$

Interpretation and Examples

Difficulty (β_i):

- Represents the **location** of the item on the latent trait scale.
- It's the value of θ_j at which the respondent has a 50% chance of a positive response.

Examples:

- A bill with $\beta_i = 0$: a moderate legislator is equally likely to vote "yes" or "no."
- A bill with $\beta_i = 2$: only more extreme right-leaning legislators will support it.

Discrimination (α_i):

- Reflects how **sensitive** the item is to differences in the latent trait.
- A high α_i produces a steep logistic curve: it clearly separates respondents near β_i .
- A low α_i results in a flatter curve: response probabilities change slowly across θ_j .

Examples:

- A symbolic or low-stakes vote (low α_i) does not sort legislators strongly by ideology.
- A pivotal vote on ideological grounds (high α_i) creates clear divisions — moderate and extreme legislators split predictably.

Political Science Analogy

In roll-call voting:

- θ_j : legislator's **ideal point** on the ideological spectrum
- β_i : the ideological **center** of a bill — how liberal/conservative it is
- α_i : **salience or divisiveness** — how much ideological distance matters

Such models underpin influential research on congressional polarization (e.g., Poole & Rosenthal's NOMINATE) and help uncover latent structures in public opinion and elite behavior.

Implementing a 2-Parameter IRT Model in **brms**

We are next going to implement a 2-Parameter IRT model, as we discussed in lecture. We are using **brms** for its flexibility and its extensibility. So, work through this code, but keep in mind the power of the method: how easy it is to extend it to count or continuous data, conduct type 1/2/3-Parameter models, incorporate covariates, different priors, and so on. Before going into this example, here's the **R** code from this [link](#):

 v100i05.R.r 17.4 KB

 v100i05.pdf 914.3 KB

If you finish the code below, return here for more ideas. But let's move onto the data. We can begin by loading in the packages that we will need.

1. Load Packages

```
library(filibustr) library(ggplot2) library(tidyverse) library(brms) library(dplyr)
```

Next, let's download the data for the 117th Congress and filter for votes that were taken.

2. Download & Filter the Raw Vote Data

```
# Download data for the 117th Senate (adjust `117` to another Congress if you like)
url <- "https://voteview.com/static/data/out/votes/S117_votes.csv"
votes_raw <- read.csv(url) # Keep only final "Yea" (1) and "Nay" (6) votes
votes_filtered <- votes_raw %>% filter(cast_code %in% c(1, 6))
```

What's happening?

1. We grab the CSV of all roll-call votes from Voteview.
2. We filter to just the *final passage* votes—encoded here as **1 = Yea** or **6 = Nay**.

After you've reshaped the roll-call data into a legislator × bill matrix, be sure to explicitly preserve rather than drop missing votes (e.g. via `values_fill = NA` in your `pivot_wider`). A missing vote is *not* the same as a Nay, and by keeping those `NA`s you let **brms** automatically treat them as unobserved when you later refilter to long form (`filter(!is.na(vote))`). This approach avoids inadvertently mis-aligning legislators or conflating abstentions with "No" votes, and it means you never have to impute or manually code "missing" by hand—**brms** simply skips over those rows in the likelihood, giving you a clean, principled IRT fit.

Next—we turn it into a normal data matrix, where each row is a legislator-bill vote:

3. Reshape into a Legislator × Bill Matrix

```
# Spread so each row is a legislator, each column a roll call
vote_matrix <- votes_filtered %>% select(icpsr, rollnumber, cast_code) %>% pivot_wider(
  names_from = rollnumber, values_from = cast_code ) %>% arrange(icpsr) # Recode
# Recode Yea/Nay to 1/0, leaving other codes (e.g. absences) as NA
vote_matrix[, -1] <- lapply(vote_matrix[, -1], function(x) ifelse(x == 1, 1, ifelse(x == 6, 0, NA)))
# Convert to long format
votes_long <- vote_matrix %>% pivot_longer(-icpsr, names_to = "rollcall", values_to = "vote") %>% filter(!is.na(vote)) %>%
  mutate(rollcall = factor(rollcall), legislator = factor(icpsr))
```

What's happening?

- `pivot_wider()` turns the long vote list into a matrix: one row per `icpsr` ID, one column per `rollnumber`.
- We then recode `1→1` (Yea), `6→0` (Nay), and everything else to `NA` (for missed or procedural votes).

For the sake of time, we are just going to look at the 50 randomly selected votes. For a full analysis, just switch `votes_short` to `votes_long` in the code below.

4. Filter to the 50 Most-Complete Roll Calls

```
# Identify the 50 bills with the most non-missing votes
top_rollcalls <- vote_matrix %>% select(-icpsr) %>% summarise(across(everything(), ~ sum(!is.na(.)))) %>% pivot_longer(everything(), names_to = "rollcall", values_to = "n_votes") %>% arrange(desc(n_votes)) %>% slice_head(n = 50) %>% pull(rollcall)
# Build a "short" long-form dataset on just those 50 roll calls
votes_short <- vote_matrix %>% select(icpsr, all_of(top_rollcalls)) %>% pivot_longer(-icpsr, names_to = "rollcall", values_to = "vote") %>% filter(!is.na(vote)) %>% mutate(rollcall = factor(rollcall), legislator = factor(icpsr), vote # stays 0/1 )
```

What's happening?

- We count how many non-`NA` votes each roll call has, pick the top 50, and then
- *Re-pivot* those 50 columns back into long form for modeling.

Next, it is time to set up the two-parameter model, using the logistic link.

5. Specify the 2-Parameter Logistic IRT Model

```
formula_2pl_nl <- bf( vote ~ inv_logit(logalpha * eta), eta ~ 1 + (1 | legislator), # Ideal point per legislator
logalpha ~ 1 + (1 | i | rollcall), # Discrimination per roll-call
nl = TRUE )
```


What's happening?

We fit a **nonlinear** 2PL IRT:

- `eta[j]` is legislator j 's ideal point (θ_n), with a Normal group prior across legislators.
- `logalpha[i]` is discrimination for bill i (α_i), again hierarchically modeled.
- The response model is

$$P(\text{Yea}_{i,j}) = \text{logit}^{-1}(\alpha_i \cdot \eta_j)$$

Prior specification. Let's just put normals over everything, just as a standard first attempt. Given that the model is composed entirely of latent variables, these assumptions are reasonable.

6. Set Weakly-Informative Priors

```
priors_2pl_nl <- c( prior(normal(0, 1), nlpar = "eta"), prior(normal(0, 1),  
class = "sd", group = "legislator", nlpar = "eta"), prior(normal(0, 1), nlpa  
r = "logalpha"), prior(normal(0, 1), class = "sd", group = "rollcall", nlpar  
= "logalpha") )
```

What's happening?

- We center both `eta` and `logalpha` at 0 with unit-scale normals.
- Their group-level `sd`'s also get Normal(0,1)+ priors, softly regularizing the variance.

7. Fit the Model in `brms`

Next, we do a short run, due to time. For better results, increase `iter = 4000` and `warmup = 2000`.

```
model_2pl_nl <- brm( formula = formula_2pl_nl, data = votes_short, family =  
bernoulli(), prior = priors_2pl_nl, chains = 4, cores = 4, iter = 400, warmu  
p = 200, control = list(adapt_delta = 0.95) ) variables(model_2pl_nl)
```

What's happening

We launch four Stan chains, each 400 iterations (200 warmup + 200 sampling).

The higher `adapt_delta` guards against divergent transitions.

8. Extract & "Align" Legislators' Ideal Points

```
# Extract & "Align" Legislators' Ideal Points post <- as_draws_df(model_2pl_
nl) # 1. Pull out the r_legislator__eta[legislator,Intercept] columns theta_
draws <- post |> # Retain .draw! select(.draw, starts_with("r_legislator__et
a[")) # 2. Pivot long, extract ICPSR from the name "r_legislator__eta[123,In
tercept]" theta_long <- theta_draws %>% pivot_longer( cols = - .draw, # Pivo
t all columns except .draw names_to = "param", values_to = "value" ) %>% mut
ate( legislator = as.integer(str_extract(param, "(?<=\\[)\\d+")) ) # 3. Beca
use ideal points are only identified up to # a sign-flip, we "align" each dr
aw to the first draw: reference_legislator <- min(theta_long$legislator) the
ta_aligned <- theta_long |> group_by(.draw) |> mutate( ref_val = value[legis
lator == reference_legislator], flip = if_else(ref_val < 0, -1, 1), value_al
igned = value * flip ) |> ungroup() # 4. Summarize theta_summary <- theta_al
igned |> group_by(legislator) |> summarize( mean = mean(value_aligned), lowe
r = quantile(value_aligned, 0.025), upper = quantile(value_aligned, 0.975) )
```

What's happening?

1. We grab each legislator's posterior draws of η_n .
2. We reshape so each row is a single draw-legislator pair.
3. *Sign indeterminacy*: IRT ideal points are only identified up to a global ± 1 flip—so we correlate each draw against the first draw and flip if it's negative.
4. Finally we compute posterior means and 95% credible intervals.

9. Join in Names & Party, and Recode

We next merge the legislators' names and party ID into the data....

What's happening?

We pull in each Senator's name and party code, then map codes to labels.

```
# Join in Names & Party, and Recode members <- get_voteview_members(chamber = "Senate", congress = 117) theta_named <- theta_summary |> left_join(members |> select(icpsr, bioname, party_code), by = c("legislator" = "icpsr")) |> rename(name = bioname, party = party_code) |> mutate( party = factor(case_when( party == 100 ~ "Democrat", party == 200 ~ "Republican", party == 328 ~ "Independent", TRUE ~ "Unknown" )) )
```

10. Plot the Ideal-Point Distributions by Party

and add a figure.

```
ggplot(theta_named, aes(x = party, y = mean, fill = party, color = party)) +
  geom_boxplot(alpha = 0.7, outlier.shape = 21, outlier.fill = "white") + geom_jitter(width = 0.1, alpha = 0.6, size = 1.5) + scale_fill_manual(values = c( "Democrat" = "#1f77b4", "Republican" = "#d62728", "Independent" = "#2ca02c", "Unknown" = "grey50" )) + scale_color_manual(values = c( "Democrat" = "#1f77b4", "Republican" = "#d62728", "Independent" = "#2ca02c", "Unknown" = "grey50" )) + labs( title = "Senators' Estimated Ideal Points by Party", subtitle = "Posterior means from a 2PL IRT model (S117)", x = "Party", y = "Posterior Mean of Ideal Point" ) + theme_minimal(base_size = 14) + theme(legend.position = "none")
```

What's happening?

- A boxplot & jitter of each party's distribution of $\hat{\eta}$.
- We use custom party colors to match conventional blue/red conventions.

Beyond simply scaling up the number of roll-calls, one natural extension is to **incorporate hierarchical shrinkage priors**—for instance, a Horseshoe or Dirichlet-Laplace prior—directly on the discrimination parameters (the “logalpha” terms). That lets the model decide, in a data-driven way, which votes are most informative about the underlying dimension and which are essentially “noise” (i.e., discrimination near zero). In practice this can dramatically improve the stability of ideal-point estimates when you have dozens—or even hundreds—of measures, many of which may carry only weak signal.

Another fruitful direction is to **model multiple dimensions of voting behavior**. Rather than forcing a single liberal-to-conservative scale, you could fit a two- or three-dimensional IRT model in **brms** by allowing legislators to have a vector of latent traits (budget, social issues, foreign policy) and fitting separate discrimination parameters for each vote on each dimension. This makes it possible to cluster bills by issue area and to see, for example, which senators are “fiscal hawks” but socially moderate, or vice versa.

If you have access to the full text of each bill or amendments, you could go further and **couple the IRT with a text-analysis component**. One approach is to featurize each roll-call statement by its word counts or topic-mixture weights, then model the discrimination parameters as a function of those text features. That way you learn not only each legislator’s ideal point, but also which kinds of language—“tax relief,” “national security,” “environmental regulation”—tend to draw more partisan divergence.

Finally, for longitudinal studies you can let **ideal points drift over time** by adding an AR(1) or random-walk prior on each legislator’s theta across sessions. This will show you who “moderated” or “radicalized” over the course of a Congress. Combined with rich covariates—committee seats, fundraising totals, district partisanship—you’ll have a fully fleshed-out model of how policy, politics, and personal constituency pressure all converge in the roll-call chamber.

 [Class Responses](#)

Research Focus: Measurement That Matches Theory: Theory-Driven Identification in Item Response Theory Models (2024) Morucci, et al. ([link](#))

In the IRT-M framework, each response “item” (e.g. a particular survey question–answer) is linked to each latent dimension by a small diagonal “constraint” matrix M_k . For item k and factor (latent dimension) j , the diagonal entry m_{kjj} is set—*before* looking at the data—to one of

- **+1** if theory says “higher values of factor j should *increase* the probability of a ‘yes’ on item k ,”
- **–1** if theory says “higher j *decrease* the probability,”
- **0** if theory says “ j plays *no* role for that item,”
- **NA** if you simply have *no prior* on whether j matters.

1. How that translates into the model

In the usual two-parameter probit IRT

$$\Pr(y_{ik} = 1) = \Phi(\lambda_k^\top \theta_i - b_k)$$

each λ_{kj} (the loading of item k on factor j) is now given a *truncated normal* prior whose center and sign are determined by m_{kjj} . Concretely,

$$\lambda_{kj} \sim \begin{cases} \mathcal{N}_{(0,\infty)}(0, m_{kjj}^2) & \text{if } m_{kjj} > 0, \\ \mathcal{N}_{(-\infty,0)}(0, m_{kjj}^2) & \text{if } m_{kjj} < 0, \\ \delta_0 & \text{if } m_{kjj} = 0, \\ \mathcal{N}(0, 5) & \text{if } m_{kjj} = \text{NA}, \end{cases}$$

so that the loading can only be positive, negative, zero, or unconstrained, exactly as your *prior theory* dictates.

2. Identification via the M matrices

A bare IRT model suffers from three indeterminacies:

1. **Location shift:** you can add a constant to all $\lambda^\top \theta$ and subtract it from the b_k without changing anything. That's fixed by standard priors on b_k .
2. **Scale flip:** you can multiply each λ by a and divide each θ by a . That's fixed by standard priors on the θ_i .
3. **Rotation:** in a d -dimensional factor model you can apply *any* $d \times d$ orthogonal rotation Q to λ and θ and get the same likelihood.

What's new in IRT-M is that by demanding *at least* $d(d-1)$ of the λ_{kj} be exactly zero (equivalently, by setting those $m_{kjj} = 0$, you break the rotation invariance. Intuitively, if item k is forced *not* to load on factor j , any rotation that mixes j into that item's loading would violate your zeros. In practice you achieve a fully identified model by combining

- **Zero-loadings:** at least $d(d-1)$ constraints $m_{kjj} = 0$, **and/or**
- **Anchor points:** synthetic "perfect extreme" respondents whose pattern of yes/no across items pins down each axis of variation.

Either route (or a combination of both) *anchors* your latent space so you don't need to fix any legislators' or respondents' positions exogenously, and you never lose the clean Bayesian Gibbs-sampling structure.

3. Substantive interpretation "for free"

Because every nonzero prior on λ_{kj} is *directly* coded from your theory, *each* latent dimension is guaranteed to correspond to a concept you chose ahead of time. If you label factor j "economic threat," then *only* the items you coded $m_{kjj} = +1$ or -1 will ever load on that dimension. No mysterious "second factor" should ever pop up with an opaque mix of social, religious, and economic content. Instead, you get

1. **Clear meaning:** "Factor 3" is by construction the *economic*-threat axis,
2. **Correct sign:** higher factor 3 *increases* "yes" on the job-security item if $m_{job,3} = +1$
3. **Zero-loading:** factor 3 *never* affects the cultural-values item if $m_{cult,3} = 0$

All of which means you've built a measurement that **exactly matches** your theory, *and* you've simultaneously tamed all of the usual IRT identification headaches.